

Homework 4 :

Ansible installation based on Docker

Highlight steps:

- docker run ansible-slave
 - Set it correctly
 - docker run ansible-master
 - Set it correctly
 - Create inventory configuration file
-

For start need to run docker ansible-slave :

Create docker slave :

```
shahaf@ShahafJr: /mnt/c/Use x + v
shahaf@ShahafJr: /mnt/c/Users/shaha/svcGitLab/ex-4/slave$ cat Dockerfile
FROM ubuntu:22.04
RUN apt update && apt install -y openssh-server
RUN mkdir /var/run/ssh
RUN useradd -m slave && echo "slave:2284" | chpasswd
EXPOSE 22
CMD ["/usr/sbin/sshd", "-D"]
shahaf@ShahafJr: /mnt/c/Users/shaha/svcGitLab/ex-4/slave$ |
```

Base on ubuntu and then need to download ssh

- For sshd (secure shell daemon) need to open directory to all the listening incoming on TCP port 22. The location in the natural location base on google.
- Need to create ansible user whit password.
- Open port 22 for the connection
- CMD – run the ssh server on the beginning, The reason is the container need to stay alive.

Now build the image with tag 'ansible-slave'

```
shahaf@ShahafJr: /mnt/c/Use x + v
shahaf@ShahafJr: /mnt/c/Users/shaha/svcGitLab/ex-4/slave$ docker build -t shahafz/ansible-slave .
[+] Building 2.3s (9/9) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 221B 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:22.04 0.8s
=> [auth] library/ubuntu:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/4] FROM docker.io/library/ubuntu:22.04@sha256:eb29ed27b0821dca09c2e28b39135e185fc13020364 0.0s
=> => resolve docker.io/library/ubuntu:22.04@sha256:eb29ed27b0821dca09c2e28b39135e185fc13020364 0.0s
=> CACHED [2/4] RUN apt update && apt install -y openssh-server 0.0s
=> CACHED [3/4] RUN mkdir /var/run/ssh 0.0s
=> CACHED [4/4] RUN useradd -m slave && echo "slave:2284" | chpasswd 0.0s
=> exporting to image 1.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:2d7508477ed5e3caf2d0084e88598b8f8ca05e88d053978250f4ce9567d81f9 0.0s
=> => exporting config sha256:24b1100a90c8353292a78e150138198d2a45ab2f16dc90767ac1b4d4e625a390 0.0s
=> => exporting attestation manifest sha256:cff64b0ce3b0e873715e2bea5a831c0178241dcedf9f19078c8 0.0s
=> => exporting manifest list sha256:5866c691ec3c4e3fbf70927bf9927a3e2bca6fdb097fe3de17b8cf2fc3 0.0s
=> => naming to docker.io/shahafz/ansible-slave:latest 0.0s
=> => unpacking to docker.io/shahafz/ansible-slave:latest 1.2s
shahaf@ShahafJr: /mnt/c/Users/shaha/svcGitLab/ex-4/slave$ |
```

upload the image to docker hub

```
shahaf@ShahafJr:/mnt/c/Use x + v
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$ docker images
INFO Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
jenkins/jenkins:lts 23b84d1066db 802MB 288MB
movie:1.00.01 8406ec9eb95d 842MB 190MB U
mysql:latest 24e450bbd24f 1.27GB 283MB U
shahafz/ansible-slave:latest 5866c691ec3c 412MB 121MB
shahafz/my-app:1.0 3fe4da414b55 200MB 48.8MB
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$ docker push shahafz/ansible-slave:latest
The push refers to repository [docker.io/shahafz/ansible-slave]
bf6dc80f43fb: Pushed
40ceac852dec: Pushed
de47083ed7d7: Pushed
5e06f5c956aa: Pushed
b6b19d7075ef: Pushed
latest: digest: sha256:5866c691ec3c4e3fbf70927bf9927a3e2bca6fdb097fe3de17b8cf2fc3897698 size: 856
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$ docker rmi shahafz/ansible-slave:latest
Untagged: shahafz/ansible-slave:latest
Deleted: sha256:5866c691ec3c4e3fbf70927bf9927a3e2bca6fdb097fe3de17b8cf2fc3897698
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$
```

Now create container slave listen on port 22.
and host can access on port 2222

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$ docker run -d -p 2222:22 shahafz/ansible-slave
Unable to find image 'shahafz/ansible-slave:latest' locally
latest: Pulling from shahafz/ansible-slave
40ceac852dec: Pull complete
5e06f5c956aa: Pull complete
b6b19d7075ef: Pull complete
bf6dc80f43fb: Download complete
Digest: sha256:5866c691ec3c4e3fbf70927bf9927a3e2bca6fdb097fe3de17b8cf2fc3897698
Status: Downloaded newer image for shahafz/ansible-slave:latest
b49f11a0cc0f46f1d2337fb8d11d6e8d839a51593dea0469d8549603e5472ae
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$ docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
b49f11a0cc0 shahafz/ansible-slave "/usr/sbin/sshd -D" 11 seconds ago Up 10 seconds 0.0.0.0:2222->22/tcp, [::]:2222->22/tcp brave_haslett
2e68ca146ff0 movie:1.00.01 "docker-entrypoint.s..." 42 hours ago Up 42 hours 0.0.0.0:5000->5001/tcp, [::]:5000->5001/tcp blissful_banzai
6db5a2a5e384 mysql "docker-entrypoint.s..." 9 days ago Exited (255) 42 hours ago 33060/tcp, 0.0.0.0:3406->3306/tcp mysql-server
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$
```

Check the ssh connected user slave and password 2284.

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/slave$ ssh -p 2222 slave@localhost
slave@localhost's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

$ |
```

Create a network - bridge

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker network create my-network
9432d0aefbb5493c88de0fd89cc61975cad83332aea9381976c9e6b66f7ce80d
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
255f21c64863        bridge              bridge              local
49be69d09f25        host                host                local
9432d0aefbb5        my-network          bridge              local
ce2689998229        none                null                local
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$
```

The information on the new network and 16 sub option inside. [172.18.0.1-16]

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker network inspect my-network
[
  {
    "Name": "my-network",
    "Id": "9432d0aefbb5493c88de0fd89cc61975cad83332aea9381976c9e6b66f7ce80d",
    "Created": "2026-04-07T14:41:22.316762719Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
  }
]
```

And connect the slave to it

Create network bridge and give him name for the reason to connect inside this network between master and slave. connect the slave to it

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker run -d --name ansible-slave --network my-network shahafz/ansible-slave
b2f92af988a9a7b19a16f64e3e556d7b2017632f38b1c7e891678d7dba6a54d0
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$
```

The network inside the ansible-slave

```
root@b49f111a0cc0: /
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker exec -it b49f111a0cc0 bash
root@b49f111a0cc0:/# hostname -I
172.17.0.3
root@b49f111a0cc0:/#
```

Connect the master to it also

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker run -d --name ansible-master --network my-network shahafz/ansible-master
29b16edec31ef417c542d7d24829c4413f87ac979ecde1aaab02272acc485b46
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$
```

check connected

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker inspect ansible-slave | grep -w "my-network"
    "NetworkMode": "my-network",
    "my-network": {
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker inspect ansible-master | grep -w "my-network"
    "NetworkMode": "my-network",
    "my-network": {
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$
```

Now for the master

Create Dockerfile:

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker build -t shahafz/ansible-master .
[+] Building 52.4s (10/10) FINISHED                                docker:default
=> [internal] load build definition from dockerfile                0.1s
=> => transferring dockerfile: 171B                                0.0s
=> [internal] load metadata for docker.io/library/ubuntu:22.04    1.0s
=> [auth] library/ubuntu:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> CACHED [1/4] FROM docker.io/library/ubuntu:22.04@sha256:eb29ed27b0821dca09c2e28b39135e185fc1302036427d5f4d70a41ce8fd7659 0.1s
=> => resolve docker.io/library/ubuntu:22.04@sha256:eb29ed27b0821dca09c2e28b39135e185fc1302036427d5f4d70a41ce8fd7659 0.1s
=> [internal] load build context                                   0.1s
=> => transferring context: 101B                                     0.0s
=> [2/4] RUN apt update && apt install -y ansible                 33.8s
=> [3/4] RUN apt install -y openssh-client                       1.3s
=> [4/4] COPY inventory inventory                                0.3s
=> exporting to image                                             15.7s
=> => exporting layers                                             9.5s
=> => exporting manifest sha256:36332567e0370d486e5dddf96467551971ad365db4efd1e4b63417e9339f098af 0.0s
=> => exporting config sha256:bbb6138ab622719c20e055c97f61c2543c2f8841d5f9715af2c1b5de9f36f73e 0.0s
=> => exporting attestation manifest sha256:17dda328a3178c60e57c800925f31d3b0594ef053da3bdc2fbc7d343f5708bde 0.0s
=> => exporting manifest list sha256:483f637aeba0da4d4242612c0b70c2a2bf311968d945970db69830ebb1487aad 0.0s
=> => naming to docker.io/shahafz/ansible-master:latest          0.0s
=> => unpacking to docker.io/shahafz/ansible-master:latest       6.0s
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ |

-----
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker run -d -p 2000:2001 shahafz/ansible-master
Unable to find image 'shahafz/ansible-master:latest' locally
latest: Pulling from shahafz/ansible-master
b5290ef1152a: Pull complete
f280874d4c14: Pull complete
756ea4619412: Pull complete
bc79ecd92257: Download complete
Digest: sha256:483f637aeba0da4d4242612c0b70c2a2bf311968d945970db69830ebb1487aad
Status: Downloaded newer image for shahafz/ansible-master:latest
3d99adba122f507161fc1a0354716fd4f3e4d3163868eaebed771a8f096e6e4b
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ |
```

Run the container and also connect the master to the network.

And run the script

```
shahaf@ShahafJr:/mnt/c/Users/shaha/svcGitLab/ex-4/master$ docker run -it --name ansible-master --network my-network shahafz/ansible-master
root@6fcb6422bd44:/app# ansible -m ping all
ansible-slave | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
root@6fcb6422bd44:/app# |
```